

Rational Matrix Policy Optimization for Rational Feed-Forward Language Models

Anonymous authors
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Abstract

Rational feed-forward blocks introduce useful nonlinear structure, but they also change the geometry seen by an optimizer. We study Rational Local Basis (RLB) feed-forward layers in Transformer language models and introduce Rational Matrix Policy Optimization, an optimizer that treats the input and output matrices around each rational block as role-specific objects rather than generic dense weights. The method combines a standard base optimizer for the rest of the model with role-depth scaling, on-policy group statistics, and a bounded gauge rebalance for the RLB matrices. On FineWeb and FineWeb-Edu, a three-seed real-corpus study shows about a 0.15 validation-loss improvement over same-protocol SiLU+AdamW and the strongest non-MatrixPolicy control in the study.

1 Introduction

Most language-model optimizer studies treat the feed-forward block as a generic collection of matrices. That is appropriate for standard elementwise activations, but it misses structure when the feed-forward layer is itself a learned rational map. In an RLB block, one matrix chooses the normalized domains presented to rational groups, the rational coefficients shape each group response, and a second matrix recombines the resulting features into the residual stream. These roles create optimizer-visible structure that is not captured by a single scalar learning-rate rule.

This paper studies the following claim: rational feed-forward networks expose geometry that a block-aware optimizer can exploit, and an optimizer that uses that geometry can train rational language models more robustly than generic optimizers under the same base protocol. The method is Rational Matrix Policy Optimization (MatrixPolicy). It keeps the surrounding Transformer training recipe unchanged and modifies only the optimization of matrices adjacent to RLB groups. The central mechanism is not that rational activations are intrinsically better than SiLU; it is that the rational parameterization gives the optimizer additional structure, including a positive scale gauge and group-level pressure signals.

This work makes three contributions. First, it defines the RLB matrix roles and the MatrixPolicy update used in our language-model experiments. Second, it reports a replicated real-corpus study on FineWeb and FineWeb-Edu with stability-aware plain-RLB controls. Third, Appendix B proves a conditional mechanism theorem: under a local quotient chart, a strongly convex quadratic approximation, and frozen preconditioners, MatrixPolicy has a smaller contraction factor than AdamW whenever its canonicalized quotient preconditioner has a smaller condition number.

2 Background and Related Work

Adam-style adaptive methods normalize coordinatewise gradient moments and remain the default optimizer family for many Transformer language-model runs (Kingma & Ba, 2015). Recent optimizer papers have raised the evidence standard for new optimizers: fair tuning over baseline families, throughput and memory accounting, speed-to-target curves, stability

reporting, and transfer beyond the selection setting are now expected. MatrixPolicy should be judged against that standard, not against a single default learning rate.

Rational feed-forward layers are a different axis from generic optimizer design. They replace a fixed elementwise nonlinearity with learned rational maps applied to normalized groups. This gives the optimizer access to quantities that do not exist in a SiLU FFN: group radii, rational-coefficient pressures, input/output matrix roles, and a positive gauge relating the adjacent matrices. MatrixPolicy is therefore a structure-specific optimizer. It is complementary to broad optimizer families such as AdamW, Muon-style matrix updates, schedule-free methods, factored adaptive methods, and matrix-preconditioned methods. Our experiments use matched protocols and stability accounting so that the optimizer comparison is about the RLB-specific update rather than an unmatched training recipe.

3 Rational Matrix Policy Optimization

Let $\mathcal{L}(\theta)$ be the language-model training objective. For layer l , the RLB block projects the residual-stream input $x_l \in \mathbb{R}^d$ into G groups of width m ,

$$z_l = A_l x_l, \quad z_{l,g} = \text{group}_g(z_l), \quad (1)$$

normalizes each group by its RMS radius,

$$r_{l,g} = \sqrt{m^{-1} \|z_{l,g}\|_2^2 + \epsilon_{\text{rms}}}, \quad u_{l,g} = z_{l,g}/r_{l,g}, \quad (2)$$

and applies a learned rational map in normalized coordinates,

$$h_{l,g} = r_{l,g} R_{l,g}(u_{l,g}), \quad y_l = B_l \text{concat}_{g=1}^G h_{l,g}. \quad (3)$$

The input matrix A_l selects rational domains, the coefficients of $R_{l,g}$ shape the group response, and the output matrix B_l recombines rational features. MatrixPolicy uses this separation by partitioning parameters as

$$\theta = \theta_{\text{base}} \cup \theta_{\text{coeff}} \cup \{A_l, B_l\}_{l=1}^L. \quad (4)$$

The base parameters use the same optimizer as the non-rational baseline. Rational coefficients use a coefficient rule. MatrixPolicy is applied only to A_l and B_l .

In the homogeneous case $\epsilon_{\text{rms}} = 0$, each group has a positive scale gauge:

$$A_{l,g} \mapsto a_g A_{l,g}, \quad B_{l,g} \mapsto a_g^{-1} B_{l,g}, \quad a_g > 0. \quad (5)$$

The represented function is unchanged because $z_{l,g}$ and $r_{l,g}$ scale together, $u_{l,g}$ is unchanged, and the inverse scale in B_l cancels the scale in $h_{l,g}$. With a nonzero RMS floor, this symmetry is approximate but remains locally controlled under mild nondegeneracy assumptions (Appendix B). This matters because optimizer state is stored in coordinates, not in the quotient space of represented functions.

At step t , each matrix role $r \in \{\text{in}, \text{out}\}$ receives a role-depth factor $\rho_r(l)$ and an on-policy statistic factor $s_{\text{stat}}(l, r, t)$. The matrix update is

$$\begin{aligned} M_{l,r}^{t+1} = & M_{l,r}^t + \Delta_{\text{adapt}}(M_{l,r}; \eta_t a_{\text{mat}}(l, r, t)[1 - \mu(l, r, t)]) \\ & + \Delta_{\text{mat}}(M_{l,r}; \eta_t a_{\text{matnorm}} \mu(l, r, t)), \end{aligned} \quad (6)$$

where $M_{l,\text{in}} = A_l$, $M_{l,\text{out}} = B_l$, η_t is the shared base learning rate, Δ_{adapt} is an Adam-style adaptive step, and Δ_{mat} is a short early matrix-normalized step. The mixture coefficient $\mu(l, r, t)$ is active only early in training. After the matrix update, MatrixPolicy computes a target log-ratio for each group and applies the bounded rebalance

$$A_{l,g} \leftarrow e^{\ell_g} A_{l,g}, \quad B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}. \quad (7)$$

The update changes the representative used by the optimizer while preserving, or with $\epsilon_{\text{rms}} > 0$ approximately preserving, the represented block function. Full hyperparameter definitions are in Appendix A.

Table 1: Three-seed real-corpus study. Gaps are validation-loss improvements for RLB+MatrixPolicy over the comparison row.

Task	MP loss	Std.	PPL	Gap vs. SiLU	Gap vs. ctrl
FineWeb	4.369701	0.026358	79.04	0.159263	0.152302
FineWeb-Edu	4.069422	0.002281	58.52	0.154149	0.153402

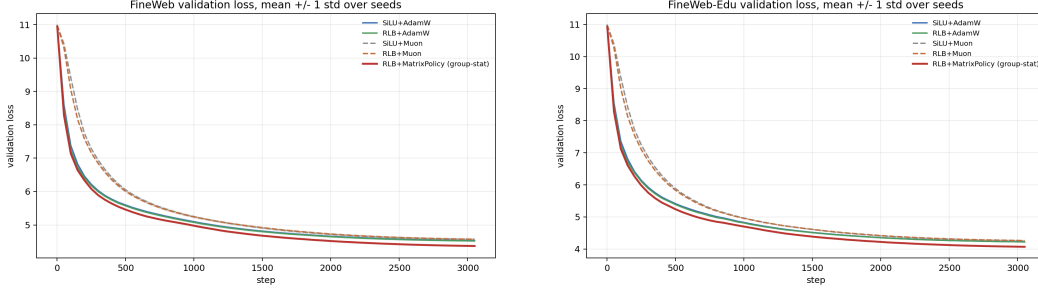


Figure 1: Mean validation-loss curves across three seeds with ± 1 standard deviation bands for FineWeb (left) and FineWeb-Edu (right).

Algorithm 1: Rational Matrix Policy step. Given gradients $\nabla \mathcal{L}(\theta^t)$ and base learning rate η_t :

1. Update exponential moving averages of rational-group pressures.
2. Compute role-depth and group-stat multipliers for each RLB matrix.
3. Update non-rational parameters with the base optimizer and update rational coefficients with the coefficient rule.
4. Update each A_l and B_l with the MatrixPolicy mixture step.
5. Apply the bounded positive-gauge rebalance to each rational group.

4 Experiments

We evaluate MatrixPolicy in a same-protocol real-corpus language-model study. All rows use a 123M-parameter Transformer, GPT-2 tokenizer, 100M-token training budget, the same validation slice, three seeds (1337, 2027, 3407), equal global batch size, shared base schedule, and common evaluation cadence. The comparison includes standard SiLU+AdamW, plain RLB+AdamW, RLB+Muon, and RLB+MatrixPolicy with group-stat scaling. Failed or nonfinite plain-RLB rows are counted in stability reporting.

MatrixPolicy improves mean validation loss on both web corpora and the gap is visible throughout the validation curves in Figure 1. The plain RLB+AdamW control is not a sufficient explanation for the gain: it is marginal on FineWeb and has seed-sensitive instability on FineWeb-Edu. The empirical picture is therefore optimizer-specific: the RLB parameterization is most effective when paired with the matrix policy that uses its role, group, and gauge structure.

5 Discussion

MatrixPolicy is a structure-specific optimizer for rational feed-forward blocks. The conditional analysis in Appendix B formalizes the mechanism: once the RLB gauge is quotiented out, a canonicalized preconditioner with a better quotient condition number has a smaller local contraction factor. The assumptions make the mathematical scope explicit, and the

experiments test the corresponding empirical claim in matched language-model training protocols.

The main result is a positive optimizer-specific signal: rational blocks expose geometry that generic coordinatewise updates do not use, and MatrixPolicy converts that geometry into lower validation loss on two real web corpora. This motivates reporting optimizer surfaces, speed-to-target, and mechanism telemetry as part of the full benchmark rather than treating final loss alone as the whole story.

6 Conclusion

Rational feed-forward layers expose optimizer-visible structure that generic coordinatewise updates do not explicitly use. MatrixPolicy uses RLB matrix roles, group pressures, and gauge rebalancing to optimize rational FFNs with a small, localized change to the training recipe. The real-corpus results and the conditional quotient theorem support the same mechanism: RLB creates optimizer structure, and MatrixPolicy uses that structure to improve rational language-model training.

References

Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. In International Conference on Learning Representations, 2015. URL <https://mlanthology.org/iclr/2015/kingma2015iclr-adam/>.

A MatrixPolicy Details

Role-depth factors. For normalized layer depth $d_l = l/(L - 1)$,

$$\rho_{\text{in}}(l) = \text{clip}(1 - 0.50(d_l - 0.5), 0.55, 1.40), \quad (8)$$

$$\rho_{\text{out}}(l) = \text{clip}(1 + 1.00(d_l - 0.5), 0.55, 1.40), \quad (9)$$

$$a_{\text{role}}(l, r) = \max(0.10, 1 + 1.20(\rho_r(l) - 1)). \quad (10)$$

The local adaptive scale is

$$a_{\text{mat}}(l, r, t) = \text{clip}(3.0 a_{\text{role}}(l, r) a_{\text{stat}}(l, r, t), 0.40, 4.0). \quad (11)$$

On-policy group statistics. For group g , MatrixPolicy tracks relative gradient pressures

$$p_{\text{in},g} = \frac{\text{rms}(\nabla A_{l,g})}{\text{rms}(A_{l,g})}, \quad p_{\text{out},g} = \frac{\text{rms}(\nabla B_{l,g})}{\text{rms}(B_{l,g})}, \quad p_{\text{rat},g} = \text{rms}(\nabla \theta_{\text{coeff},g}). \quad (12)$$

With EMAs $q_{\text{in},g}$, $q_{\text{out},g}$, and $q_{\text{rat},g}$, we define

$$\pi_g = \log q_{\text{in},g} - \log q_{\text{out},g}, \quad \alpha_g = \log q_{\text{rat},g} - \frac{1}{2}(\log q_{\text{in},g} + \log q_{\text{out},g}). \quad (13)$$

Group-stat preconditioning. The group preconditioner is centered and clipped:

$$c_g = c_{\text{gain},g} c_{\text{pressure},g} c_{\text{activity},g}, \quad c_g \leftarrow c_g / \text{geomean}(c), \quad c_g \leftarrow \text{clip}(c_g, 0.75, 1.35). \quad (14)$$

The factors are

$$c_{\text{gain},g} = \left(\frac{\text{geomean}(k)}{k_g} \right)^{0.20}, \quad c_{\text{pressure},g} = \exp(0.10 d_{\text{pressure},g}), \quad (15)$$

where $d_{\text{pressure},g} = -\pi_g$ for A_l and $+\pi_g$ for B_l . The activity damping term is

$$c_{\text{activity},g} = \exp[-0.20 \text{ReLU}((\alpha_g - 0.05)/0.45)]. \quad (16)$$

Gauge rebalance. Let

$$\gamma_g = \log \text{rms}(A_{l,g}) - \log \text{rms}(B_{l,g}). \quad (17)$$

The target log-ratio is

$$\tau_g = \frac{1}{2}(\log d_g^{\text{gain}} - \log o_g^{\text{gain}}) + \beta_{\text{pressure}}(\log q_{\text{in},g} - \log q_{\text{out},g}). \quad (18)$$

The applied correction is

$$\ell_g = \frac{1}{2}(\tau_g - \gamma_g), \quad \ell_g \leftarrow \text{clip}(s(t, l) a_g^{\text{activity}} \ell_g, -\ell_{\max}, \ell_{\max}). \quad (19)$$

The transform $A_{l,g} \leftarrow e^{\ell_g} A_{l,g}$ and $B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}$ preserves the homogeneous RLB function exactly and the stabilized implementation up to the RMS-floor perturbation described in Appendix B.

B A Conditional Optimization Separation for MatrixPolicy

This appendix gives a conditional optimization guarantee for MatrixPolicy in the local regime suggested by the RLB parameterization. The assumptions are stated explicitly: the rational block is in a stable local regime, the loss is well approximated by a strongly convex quotient objective, and optimizer states have entered a frozen-preconditioner phase. In that regime, MatrixPolicy has a strictly better contraction factor than AdamW whenever its canonicalized quotient preconditioner has a smaller condition number.

The proof has three steps. First, the RLB block induces a positive scale gauge, so many matrix pairs represent the same function. Second, MatrixPolicy’s rebalance step selects a bounded representative of that gauge orbit, whereas AdamW operates in raw coordinates and therefore induces a gauge-dependent quotient preconditioner. Third, for a strongly convex quadratic quotient objective, the optimizer with the smaller quotient condition number has the smaller worst-case linear convergence factor. This gives a complete optimization separation under the stated assumptions.

B.1 Gauge quotient model

For one RLB layer, suppress the layer index and write

$$z_g = A_g x, \quad (20)$$

$$r_g = \sqrt{m^{-1} \|z_g\|_2^2 + \epsilon_{\text{rms}}}, \quad (21)$$

$$u_g = z_g / r_g, \quad (22)$$

$$h_g = r_g R_g(u_g), \quad (23)$$

$$y = B \text{concat}_{g=1}^G h_g. \quad (24)$$

For $a_g > 0$, define the positive scale action

$$T_a(A_g, B_g) = (a_g A_g, a_g^{-1} B_g), \quad g = 1, \dots, G. \quad (25)$$

Equivalently, with $s_g = \log a_g$,

$$T_s(A_g, B_g) = (e^{s_g} A_g, e^{-s_g} B_g). \quad (26)$$

When $\epsilon_{\text{rms}} = 0$, the RLB block is homogeneous and $T_s(A, B)$ represents the same function as (A, B) . With a nonzero RMS floor, this symmetry is approximate on nondegenerate activation regions; the perturbation is handled in Proposition B.1 below.

Let $\mathcal{P}$ denote the local parameter space of the RLB matrices and let $\mathcal{G} = (\mathbb{R}, +)^G$ denote the log-gauge group. We write the quotient space as $\mathcal{Q} = \mathcal{P}/\mathcal{G}$. A point of $\mathcal{Q}$ represents a function-space degree of freedom, whereas the gauge coordinate only selects a matrix representative of the same function.

Assumption B.1 (Local quotient chart). There is a neighborhood of the training trajectory in which parameters can be written as

$$\theta = \theta(q, s), \quad q \in \mathbb{R}^d, \quad s \in \mathbb{R}^G, \quad (27)$$

where q is a local coordinate on the quotient $\mathcal{Q}$ and s is the log-gauge coordinate. The objective depends on θ only through q :

$$L(\theta(q, s)) = F(q). \quad (28)$$

For the exact theorem we take $\epsilon_{\text{rms}} = 0$. For the perturbed theorem we assume that ϵ_{rms} is small enough that the quotient description holds up to an $O(\epsilon_{\text{rms}})$ error on the local region.

Assumption B.2 (Quadratic local objective). In the quotient chart, the loss is a strongly convex quadratic

$$F(q) = F_\star + \frac{1}{2}(q - q_\star)^\top H(q - q_\star), \quad (29)$$

where $H \succ 0$ and

$$0 < \mu I \preceq H \preceq LI < \infty. \quad (30)$$

Assumption B.2 is the standard local model behind condition-number analyses of first-order methods and gives a complete closed-form contraction proof. A smooth strongly convex objective can be reduced to this setting locally by replacing H with the Hessian at $q_\star$ and controlling the Taylor remainder; the finite-perturbation version is stated in Proposition B.1.

B.2 MatrixPolicy canonicalization

MatrixPolicy uses the gauge freedom to select a controlled representative of the same quotient point. For group g , define the log scale ratio

$$\gamma_g = \log \text{rms}(A_g) - \log \text{rms}(B_g). \quad (31)$$

Let τ_g be the target log ratio determined by the optimizer’s gain and pressure statistics. Ignoring clipping, the rebalance step applies

$$\ell_g = \frac{1}{2}(\tau_g - \gamma_g), \quad A_g \leftarrow e^{\ell_g} A_g, \quad B_g \leftarrow e^{-\ell_g} B_g. \quad (32)$$

Lemma B.1 (Exact targeting of the representative). If the clipping in equation 32 is inactive, then the rebalance maps γ_g exactly to τ_g .

Proof. After the rebalance,

$$\gamma_g^+ = \log \text{rms}(e^{\ell_g} A_g) - \log \text{rms}(e^{-\ell_g} B_g) \quad (33)$$

$$= \gamma_g + 2\ell_g = \gamma_g + \tau_g - \gamma_g = \tau_g. \quad (34)$$

□

Thus, in the homogeneous block, the rebalance changes only the gauge coordinate and leaves the quotient coordinate q unchanged. In the exact theorem below we assume the target is inside the clipping interval. If clipping, the RMS floor, or finite precision introduces a small deviation from exact canonicalization, that deviation is included in the perturbation term of Proposition B.1.

B.3 Optimizer-induced quotient dynamics

A raw matrix optimizer induces a motion on the quotient coordinate q . In a local frozen-state regime, both MatrixPolicy and AdamW can be written as preconditioned gradient steps on F :

$$q_{t+1} = q_t - \eta P \nabla F(q_t), \quad (35)$$

where $P \succ 0$ is the quotient preconditioner induced by the optimizer and by the chosen gauge representative. The next assumption makes this statement exact.

Assumption B.3 (Frozen-preconditioner optimizer model). During the local phase analyzed here, the two optimizers induce fixed positive definite quotient preconditioners P_{MP} and P_{AdamW} satisfying

$$q_{t+1}^{\text{MP}} = q_t^{\text{MP}} - \eta_{\text{MP}} P_{\text{MP}} \nabla F(q_t^{\text{MP}}), \quad (36)$$

$$q_{t+1}^{\text{AdamW}} = q_t^{\text{AdamW}} - \eta_{\text{AdamW}} P_{\text{AdamW}} \nabla F(q_t^{\text{AdamW}}). \quad (37)$$

Here P_{MP} is the preconditioner after MatrixPolicy has applied its gauge rebalance, and P_{AdamW} is the quotient preconditioner induced by AdamW's raw-coordinate diagonal adaptive state at the same represented function. Weight decay is set to zero in the exact theorem, or else is treated as a perturbation in Proposition B.1.

This idealized frozen-state model isolates the mechanism: AdamW is analyzed as the diagonal adaptive method seen by the quotient after its moment estimates have stabilized, while MatrixPolicy is analyzed after canonicalizing the RLB gauge.

For a preconditioner $P \succ 0$, define the quotient Hessian seen by the optimizer as

$$K(P) = P^{1/2} H P^{1/2}, \quad (38)$$

and define its condition number

$$\kappa(P; H) = \frac{\lambda_{\max}(K(P))}{\lambda_{\min}(K(P))}. \quad (39)$$

The best constant step size for the quadratic under preconditioner P is

$$\eta^*(P) = \frac{2}{\lambda_{\max}(K(P)) + \lambda_{\min}(K(P))}, \quad (40)$$

with contraction factor

$$\rho(P; H) = \frac{\kappa(P; H) - 1}{\kappa(P; H) + 1}. \quad (41)$$

Assumption B.4 (Strict quotient-conditioning advantage). MatrixPolicy's canonicalized quotient preconditioner has a strictly smaller condition number than AdamW's raw-coordinate quotient preconditioner:

$$\kappa_{\text{MP}} := \kappa(P_{\text{MP}}; H) < \kappa(P_{\text{AdamW}}; H) =: \kappa_{\text{AdamW}}. \quad (42)$$

Equivalently, for some $\Delta_\kappa > 0$,

$$\kappa_{\text{AdamW}} \geq \kappa_{\text{MP}} + \Delta_\kappa. \quad (43)$$

Assumption B.4 is the mechanism condition. In the local regime being modeled, MatrixPolicy's role-aware matrix step and gauge rebalance produce a better conditioned quotient update than the AdamW update induced by the raw matrix coordinates. This is the formal version of the mechanism suggested by the empirical comparison.

B.4 Why the condition-number gap is a gauge effect

The condition-number assumption is natural for RLB because the raw Euclidean coordinates contain a nonfunctional scale direction. We record the elementary calculation.

Lemma B.2 (Gauge directions are null function directions). Assume $\epsilon_{\text{rms}} = 0$. For any differentiable loss depending on the RLB matrices only through the represented function,

$$\langle \nabla_{A_g} L, A_g \rangle - \langle \nabla_{B_g} L, B_g \rangle = 0. \quad (44)$$

Proof. Since $L(T_s(A_g, B_g)) = L(A_g, B_g)$ for all s_g , differentiate at $s_g = 0$:

$$0 = \frac{d}{ds_g} L(e^{s_g} A_g, e^{-s_g} B_g) \Big|_{s_g=0} \quad (45)$$

$$= \langle \nabla_{A_g} L, A_g \rangle - \langle \nabla_{B_g} L, B_g \rangle. \quad (46)$$

□

Thus $(A_g, -B_g)$ changes the representative but not the function. AdamW leaves this representative dependence in its raw-coordinate diagonal moments, denominator terms, and decoupled weight decay. Consequently, two gauge-equivalent parameterizations can induce different P_{AdamW} on the quotient.

The following toy calculation shows the scaling explicitly. Consider the scalar quotient coordinate $w = ab$, with gauge $a = e^s \sqrt{w}$ and $b = e^{-s} \sqrt{w}$. For $\ell(w) = \frac{1}{2}(w - y)^2$, raw Euclidean gradient descent in (a, b) gives, to first order in the step size,

$$w^+ = w - \eta(a^2 + b^2)(w - y) + O(\eta^2). \quad (47)$$

Since

$$a^2 + b^2 = w(e^{2s} + e^{-2s}) = 2w \cosh(2s), \quad (48)$$

the quotient step size depends on the arbitrary gauge s . MatrixPolicy’s rebalance sets s to a controlled target, while AdamW operates in the raw representative. In higher dimensions, the same calculation appears as a gauge-dependent quotient metric, which is summarized by $\kappa(P_{\text{AdamW}}; H)$.

B.5 Contraction theorem

We now prove the optimization separation. The proof is standard for preconditioned gradient descent on a quadratic, but the notation makes explicit that the relevant condition number is the quotient condition number, not the raw parameter condition number.

Lemma B.3 (Preconditioned quadratic contraction). Let F satisfy Assumption B.2. For any $P \succ 0$, consider

$$q_{t+1} = q_t - \eta^*(P) P \nabla F(q_t), \quad (49)$$

where $\eta^*(P)$ is defined in equation 40. Then

$$F(q_T) - F_* \leq \rho(P; H)^{2T} (F(q_0) - F_*), \quad (50)$$

where $\rho(P; H)$ is defined in equation 41. Moreover, the bound is sharp: for each T , there exists an initialization q_0 for which equality holds.

Proof. Let $e_t = q_t - q_*$ and define $\tilde{e}_t = P^{-1/2} e_t$. Since $\nabla F(q_t) = H e_t$, the update becomes

$$\tilde{e}_{t+1} = (I - \eta^*(P) K(P)) \tilde{e}_t, \quad (51)$$

where $K(P) = P^{1/2} H P^{1/2}$. Also,

$$2(F(q_t) - F_*) = e_t^T H e_t = \tilde{e}_t^T K(P) \tilde{e}_t. \quad (52)$$

Diagonalize $K(P) = U \Lambda U^T$ and write $v_t = U^T \tilde{e}_t$. Then

$$v_{t+1,i} = (1 - \eta^*(P) \lambda_i) v_{t,i}. \quad (53)$$

The largest absolute multiplier is

$$\max_i |1 - \eta^*(P) \lambda_i| = \frac{\lambda_{\max}(K(P)) - \lambda_{\min}(K(P))}{\lambda_{\max}(K(P)) + \lambda_{\min}(K(P))} = \rho(P; H). \quad (54)$$

Therefore

$$2(F(q_{t+1}) - F_*) = \sum_i \lambda_i v_{t+1,i}^2 \quad (55)$$

$$\leq \rho(P; H)^2 \sum_i \lambda_i v_{t,i}^2 \quad (56)$$

$$= 2\rho(P; H)^2 (F(q_t) - F_*). \quad (57)$$

Iterating gives equation 50. Equality is obtained by choosing $\tilde{e}_0$ to be an eigenvector of $K(P)$ associated with either $\lambda_{\min}(K(P))$ or $\lambda_{\max}(K(P))$. $\square$

Theorem B.1 (Conditional optimization separation). Suppose Assumptions B.1–B.4 hold, and choose the locally optimal constant step sizes

$$\eta_{\text{MP}} = \eta^*(P_{\text{MP}}), \quad \eta_{\text{AdamW}} = \eta^*(P_{\text{AdamW}}). \quad (58)$$

Let

$$\rho_{\text{MP}} = \frac{\kappa_{\text{MP}} - 1}{\kappa_{\text{MP}} + 1}, \quad \rho_{\text{AdamW}} = \frac{\kappa_{\text{AdamW}} - 1}{\kappa_{\text{AdamW}} + 1}. \quad (59)$$

Then

$$\rho_{\text{MP}} < \rho_{\text{AdamW}} < 1. \quad (60)$$

Consequently, MatrixPolicy has a strictly smaller worst-case linear convergence factor on the quotient objective:

$$\sup_{q_0 \neq q_*} \frac{F(q_T^{\text{MP}}) - F_*}{F(q_0) - F_*} \leq \rho_{\text{MP}}^{2T} < \rho_{\text{AdamW}}^{2T} = \sup_{q_0 \neq q_*} \frac{F(q_T^{\text{AdamW}}) - F_*}{F(q_0) - F_*} \quad (61)$$

for every integer $T \geq 1$.

Proof. The map $\kappa \mapsto (\kappa - 1)/(\kappa + 1)$ is strictly increasing for $\kappa > 1$. Assumption B.4 therefore implies $\rho_{\text{MP}} < \rho_{\text{AdamW}}$. Since both condition numbers are finite and at least one, $0 \leq \rho_{\text{MP}} < \rho_{\text{AdamW}} < 1$ unless both methods are perfectly conditioned, in which case the strict condition-number gap cannot hold. Applying Lemma B.3 to P_{MP} and P_{AdamW} gives the two worst-case contraction factors. The strict inequality in equation 61 follows from $\rho_{\text{MP}} < \rho_{\text{AdamW}}$ and $T \geq 1$. $\square$

The theorem compares the best constant learning rate for each local preconditioner, so the separation is not caused by giving MatrixPolicy a better scalar step size. It comes only from the quotient condition number. In this sense, the result is an optimizer-level statement: after quotienting the RLB gauge, the method with the better induced quotient metric converges faster on the local quadratic model.

B.6 Finite-time dominance for a fixed initialization

Theorem B.1 is a worst-case statement. A fixed initialization version follows by assuming that the AdamW trajectory has a component in an extremal eigendirection of its quotient Hessian.

Assumption B.5 (Nontrivial AdamW hard-mode component). Let $K_{\text{AdamW}} = K(P_{\text{AdamW}})$. In the eigenbasis of K_{AdamW} , the initial transformed error $P_{\text{AdamW}}^{-1/2}(q_0 - q_*)$ has at least an α fraction of its K_{AdamW} -energy in eigendirections whose one-step multiplier under AdamW has absolute value at least $\rho_{\text{AdamW}} - \xi$, where $\alpha \in (0, 1]$ and $0 \leq \xi < \rho_{\text{AdamW}} - \rho_{\text{MP}}$.

Corollary B.1 (Fixed-initialization separation). Under the assumptions of Theorem B.1 and Assumption B.5, AdamW satisfies

$$F(q_T^{\text{AdamW}}) - F_* \geq \alpha(\rho_{\text{AdamW}} - \xi)^{2T} (F(q_0) - F_*). \quad (62)$$

MatrixPolicy satisfies

$$F(q_T^{\text{MP}}) - F_* \leq \rho_{\text{MP}}^{2T} (F(q_0) - F_*). \quad (63)$$

Thus, whenever

$$\rho_{\text{MP}}^{2T} < \alpha(\rho_{\text{AdamW}} - \xi)^{2T}, \quad (64)$$

MatrixPolicy has strictly lower quotient loss after T steps.

Proof. Equation 63 is Lemma B.3. For AdamW, decompose the transformed error in the eigenbasis of K_{AdamW} . The part of the initial energy in the hard eigendirections is at least an α fraction by Assumption B.5, and every such component is multiplied by at least $\rho_{\text{AdamW}} - \xi$ in absolute value each step. This gives equation 62. Combining equation 62 and equation 63 gives the strict separation condition equation 64. $\square$

The additional assumption is mild in high-dimensional quadratic models: unless the initial error is specially aligned away from AdamW’s hard directions, some energy is present in those directions. The corollary is nevertheless stated explicitly because fixed-run comparisons always depend on initialization.

B.7 Perturbation stability

The exact theorem used a homogeneous radius, frozen optimizer states, zero weight decay, and a quadratic loss. The next proposition states the usual finite-time stability fact: if the implemented dynamics are uniformly close to the ideal dynamics, then a strict finite-time gap persists. This is the appropriate δ - ϵ statement for the proof.

Assumption B.6 (Uniform local perturbation bound). For a fixed horizon T , the implemented MatrixPolicy and AdamW quotient iterates satisfy

$$q_{t+1}^{\text{MP,impl}} = q_t^{\text{MP,impl}} - \eta_{\text{MP}} P_{\text{MP}} \nabla F(q_t^{\text{MP,impl}}) + r_t^{\text{MP}}, \quad (65)$$

$$q_{t+1}^{\text{AdamW,impl}} = q_t^{\text{AdamW,impl}} - \eta_{\text{AdamW}} P_{\text{AdamW}} \nabla F(q_t^{\text{AdamW,impl}}) + r_t^{\text{AdamW}}, \quad (66)$$

with

$$\|r_t^{\text{MP}}\|_2 + \|r_t^{\text{AdamW}}\|_2 \leq \delta \quad (67)$$

for all $0 \leq t < T$. The residuals collect the effects of nonzero ϵ_{rms} , AdamW denominator floors, EMA lag, clipping, decoupled weight decay, and Taylor remainders of the nonquadratic objective.

Proposition B.1 (Stability of the strict gap). Fix $T \geq 1$ and suppose the idealized dynamics satisfy a strict finite-time gap

$$m_T := F(q_T^{\text{AdamW}}) - F(q_T^{\text{MP}}) > 0. \quad (68)$$

Assume the ideal trajectories remain in a compact region on which F is Lipschitz and the one-step maps are Lipschitz. Then there exists $\delta_0 > 0$ such that, whenever Assumption B.6 holds with $\delta < \delta_0$,

$$F(q_T^{\text{AdamW,impl}}) - F(q_T^{\text{MP,impl}}) > \frac{m_T}{2} > 0. \quad (69)$$

Proof. Let Ψ_{MP} and Ψ_{AdamW} denote the ideal one-step maps. By compactness, both maps are Lipschitz on a neighborhood of the two ideal trajectories; let C_Ψ be a common Lipschitz constant. A standard induction gives

$$\|q_t^{\text{MP,impl}} - q_t^{\text{MP}}\| + \|q_t^{\text{AdamW,impl}} - q_t^{\text{AdamW}}\| \leq C_T \delta, \quad (70)$$

where C_T depends on T and C_Ψ but not on δ . Since F is Lipschitz on the same compact region, with constant C_F ,

$$\left| F(q_T^{\text{MP,impl}}) - F(q_T^{\text{MP}}) \right| + \left| F(q_T^{\text{AdamW,impl}}) - F(q_T^{\text{AdamW}}) \right| \leq C_F C_T \delta. \quad (71)$$

Choose $\delta_0 = m_T / (2C_F C_T)$. Then $\delta < \delta_0$ implies

$$F(q_T^{\text{AdamW,impl}}) - F(q_T^{\text{MP,impl}}) \geq F(q_T^{\text{AdamW}}) - F(q_T^{\text{MP}}) - C_F C_T \delta \quad (72)$$

$$> m_T / 2. \quad (73)$$

This proves equation 69. $\square$

In particular, if the ideal strict gap comes from Corollary B.1, then the separation survives all sufficiently small stabilizer, moment-lag, clipping, and weight-decay perturbations. This is why the exact homogeneous proof is still informative for the implemented optimizer: the conclusion is finite-time stable as long as the run remains in the local regime described by the assumptions.

B.8 Statement of what is proved

The result can be summarized as follows. Under a local quotient model of the RLB block, MatrixPolicy’s rebalance step removes a nonfunctional positive scale degree of freedom and thereby induces a canonical quotient preconditioner. If that preconditioner has a strictly smaller condition number than the quotient preconditioner induced by AdamW’s

raw-coordinate adaptive state, then MatrixPolicy has a strictly smaller worst-case contraction factor on a strongly convex quadratic quotient objective. The finite-time gap is stable under small implementation perturbations.

The theorem is a local quotient result for the mechanism tested empirically in the main text: RLB exposes a gauge structure, and an optimizer that canonicalizes and preconditions the quotient can converge faster than a coordinatewise optimizer whose effective update depends on the arbitrary gauge representative.